

TAIL-EVENT INTELLIGENCE FOR DEFI

From correlated market shocks to evidence-backed contract audits.

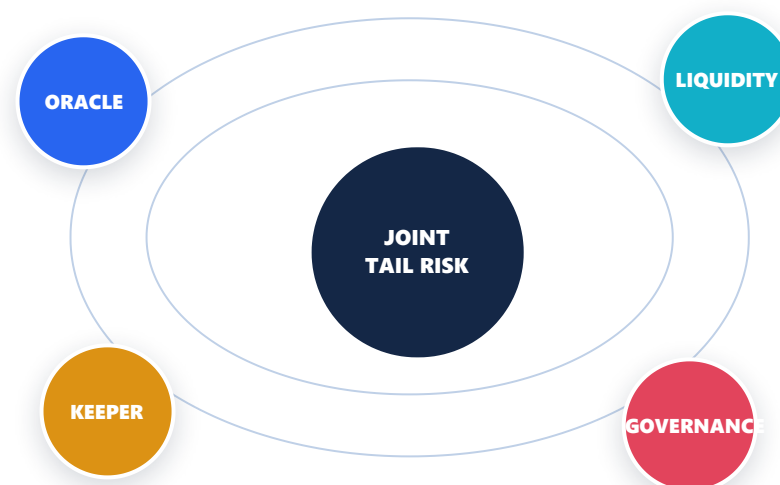
An integrated research dashboard for estimating multi-factor tail probability, testing liquidation resilience, and running traceable smart-contract audit agents.

Ethereum-first

Calibrated probabilities

Autonomous audit pipeline

Built for protocol teams, risk researchers, auditors, and investors who need more than a single risk score: they need the scenario, the dependence structure, the evidence, and the limitations.



THE PROBLEM

DeFi failures rarely arrive one factor at a time.

Liquidation crises emerge from linked mechanisms: price feeds lag, collateral gaps down, DEX depth disappears, gas rises, keepers stall, and governance controls become part of the response surface.

Fragmented analytics

Market risk and code risk are usually separated

Probability dashboards ignore contract paths; audit reports rarely quantify joint market conditions.

False precision

A single score hides the mechanism

Users need factor marginals, dependence, confidence intervals, model versions, and scenario assumptions.

Untraceable AI

LLMs can overclaim without evidence boundaries

AI should classify and explain supplied evidence, not invent probabilities, source locations, or exploitability.

OUR ANSWER

Statistical tail engine

Calibrated marginal probabilities plus factor dependence and Gaussian-copula Monte Carlo.

Contract-aware scenario builder

Search an Ethereum contract, map its protocol type to relevant factors, and select 1d / 7d / 30d horizons.

Evidence-backed audit agent

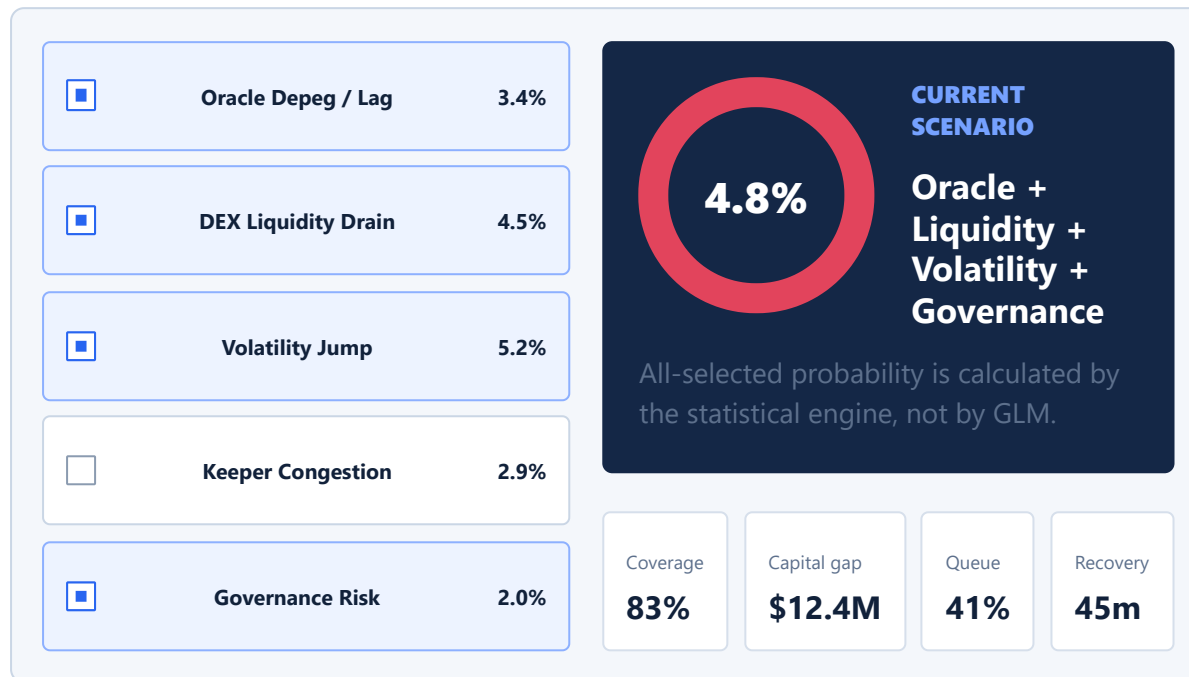
Verified source, proxy resolution, Slither, deterministic triage, GLM analyst review, and downloadable reports.

Forecast horizons	Tail factors	Historical feature rows	Audit stages
3	8	1,956	9
1d · 7d · 30d	Market + operations +	Current calibration dataset	Evidence to versioned report

INTERACTIVE WORKFLOW

Build a linked tail-event scenario.

Users choose a contract, prediction horizon, shock severity, and any combination of risk factors. Reset returns every scenario metric to zero.



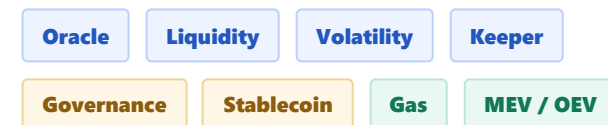
PROBABILITY DEFINITIONS

Three distinct questions, kept separate

P(all selected factors within horizon)
P(any selected factor within horizon)
P(at least two selected factors within horizon)

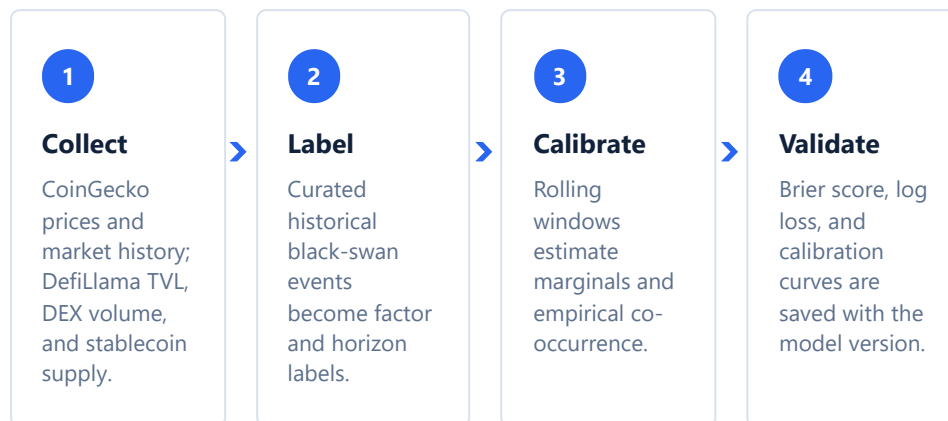
- Marginal probabilities are estimated independently for each factor and horizon.
- Historical co-occurrence is converted into a shrunk dependence matrix.
- A stabilized Gaussian copula simulates correlated factor states.
- Monte Carlo seeds make repeated calculations reproducible.
- Confidence intervals communicate sampling uncertainty instead of hiding it.

Risk factors



DATA PIPELINE

Historical signals become versioned model parameters.



CURRENT VERSION

empirical-tail-v0.3.0-20260613

Status: early calibration. The dashboard exposes the version and warnings instead of presenting exploratory estimates as settled truth.

Calibration warning

The curated event catalog contains few independent incidents and overlapping horizon labels are serially dependent. Validation currently supports exploratory research, not an unconditional guarantee of future loss frequency.

EXAMPLE 1-DAY MARGINALS



DEPENDENCE SURFACE

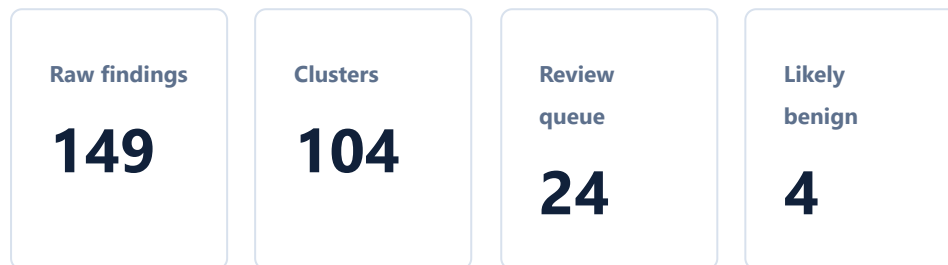
Factor	Oracle	Liquidity	Volatility	Keeper	Gas
Oracle	1.00	linked	linked	weak	weak
Liquidity	linked	1.00	strong	linked	weak
Volatility	linked	strong	1.00	linked	weak
Keeper	weak	linked	linked	1.00	strong

ONE-CLICK FULL AUDIT

From an Ethereum address to a traceable review queue.



AAVE V3 POOL EXAMPLE



High

Upgrade initialization candidate
 Source excerpt, proxy context, authorization preconditions, and recommended fork test are kept together.

Manual review

High

Explicit user transferFrom pattern
 Known protocol asset-flow patterns are downgraded but never silently removed from the evidence trail.

Likely benign

REPORT CONTENTS

Every final report is versioned and downloadable

- Evidence bundle ID and SHA-256 evidence hash
- Compiler and Slither versions
- Finding clusters with file, line, and source excerpts
- Deterministic verdict and GLM critic verdict
- Exploit preconditions and recommended tests
- Cross-factor mechanism chains
- Immediate actions and explicit limitations
- Final report hash for traceability

END-TO-END DESIGN

Each component does the job it can defend.

01

Data Sources

- Ethereum RPC
- Sourcify
- Etherscan
- CoinGecko
- DefiLlama
- Curated tail events

02

Deterministic Engines

- Contract metadata resolver
- Risk factor mapping
- Rolling marginals
- Dependence matrix
- Copula Monte Carlo
- Slither triage

03

GLM Agent Layer

- Contract classification
- Factor recommendation
- Scenario explanation
- Audit analyst review
- Adversarial critic review
- Remediation narrative

04

User Experience

- English / Chinese UI
- Contract search
- Scenario controls
- Probability heatmap
- Audit review queue
- JSON report download

Statistics owns numbers

Probabilities, confidence intervals, correlations, validation metrics, and Monte Carlo outputs come from code and data.

GLM owns interpretation

The model classifies, summarizes, challenges, and explains evidence supplied by deterministic systems.

Humans own confirmation

Exploitability, severity acceptance, remediation approval, and deployment decisions remain accountable human judgments.

HOW TO USE

One dashboard, two complementary workflows.

Tail-risk research

- Search or select an Ethereum smart contract.
- Choose 1d, 7d, or 30d.
- Select linked factors manually or ask GLM for recommendations.
- Adjust severity and dependence settings.
- Read joint probability, confidence, bad-debt, queue, and recovery estimates.

Contract audit

- Click **Run Full Audit**.
- Inspect evidence-backed candidates and source excerpts.
- Review GLM analyst and critic reasoning.

Product direction

The next maturity layer is dynamic validation: mainnet-fork exploit tests, deployed-state invariant checks, richer incident labels, and broader protocol coverage. The current product already provides the evidence and model-version foundation required to add those capabilities responsibly.

TRUST BOUNDARIES

Designed to provide

- Reproducible scenario calculations
- Versioned model and evidence metadata
- Transparent uncertainty and warnings
- Prioritized audit candidates
- Traceable AI interpretation

Does not claim

- Guaranteed future loss frequency
- Industry-certified correlations
- Automatic proof of exploitability
- Complete formal verification
- Investment or security guarantees

LOCAL LAUNCH

```
cd "E:\DeFi\Tail-Factor"
node server.js
```

Then open **http://localhost:3000**. API keys remain in PowerShell environment variables and are never exposed to the frontend.